

NSCS210 / CSC210 — Project 2

Custom UNIX Utilities — sh_ls Implementation

Group	Group 17
Repository	ncsc210-os-lab-g17
Branch	custom_ls (merged to main)
Command	sh_ls

1. Overview

The **sh_ls** command is a lightweight, custom implementation of the standard UNIX **ls** utility, written from scratch in C. It lists the contents of a specified directory (or the current directory by default), excluding hidden files. It is integrated into the Group 17 custom shell and built via the shared Makefile.

2. Implementation Details

2.1 Source File

Location: **G17_Project2_1/src/sh_ls.c**

2.2 How It Works

- Opens the target directory using `opendir()`
- Iterates over entries with `readdir()`
- Skips hidden files (entries starting with `.`)
- Prints each visible entry name and closes the directory

2.3 Key System Calls / APIs Used

Function	Purpose
<code>opendir(path)</code>	Opens a directory stream for the given path
<code>readdir(dir)</code>	Reads the next entry from the directory stream
<code>closedir(dir)</code>	Closes the directory stream after use
<code>perror()</code>	Prints error message if directory cannot be opened

3. Usage

3.1 Syntax

```
sh_ls [directory]
```

3.2 Examples

List current directory:

```
g17-shell> sh_ls
```

List a specific directory:

```
g17-shell> sh_ls /home
```

4. Build Instructions

From the project root directory:

make

Launch the shell:

```
./bin/shell
```

The `sh_ls` binary is added to TARGETS in the Makefile and compiled automatically by the pattern rule `bin/%: src/%.c`.

5. Execution Screenshot

The following screenshot shows `sh_ls` running successfully inside the Group 17 custom shell, listing the project directory contents:

